Arkaprava Saha

Curriculum Vitae

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Webpage

Education

2018–Present **Doctor of Philosophy (PhD) in Computer Science and Engineering**, *Nanyang Technological University (NTU)*, Singapore, Data Management and Analytics Laboratory.
Supervisor: Prof. Arijit Khan. CGPA: 4/5 (till date)

2014–2018 **Bachelor of Technology (BTech) in Computer Science and Engineering**, *Indian Institute of Technology (IIT) Delhi*, New Delhi, India, Departmental Specialization in Data Analytics and Artificial Intelligence.
CGPA: 8.187/10, DGPA: 8.223/10

2012–2014 **High School**, *D.A.V. Model School*, Durgapur, West Bengal, India, Affiliated to CBSE.
AISSCE (Final Exam) Grade: 95.8%

2001–2012 **Secondary School**, *St. Xavier's School*, Durgapur, West Bengal, India, Affiliated to CISCE.
ICSE (Final Exam) Grade: 95.43%

Research Interests

- Machine Learning
- Data Analytics

Publications

- **Reachability in Large Graphs Using Bloom Filters**. Arkaprava Saha, Neha Sengupta, Maya Ramanath. *IEEE International Conference on Data Engineering (**ICDE**) 2019 Workshop on Large Scale Graph Data Analytics*

Bachelors Thesis

Random Walks on Bloom Graphs

Supervisor **Prof. Maya Ramanath**

Description This project implements random walks on large Bloom graphs, whose adjacency lists are stored as lists of **Bloom filters**. This will help to answer reachability queries on large graphs like those of social networks with a much smaller memory footprint and a much lower update time than it was possible before.

Challenge Bloom filters typically have non-zero probability of **false positives** (predicting that an element is present in the set when it is actually not) and hence the **accuracy** of the random walk generated is a major issue to deal with. Moreover, the space-efficient storage of large graphs and the time-efficient generation of random walks on such graphs are some other challenging tasks.

Role My role in this project was to implement random walks on both static and dynamic bloom graphs, along with a faster and more accurate **multi-threaded algorithm** for sampling from a Bloom Filter using a **Bloom Sample Tree**.

Graduate Courses Taken

IIT Delhi Machine Learning, Logic For Computer Science, Special Topics in Machine Learning (Data Mining), Natural Language Processing, Numerical Algorithms

NTU Deep Learning for Data Science, Advanced Topics in Distributed Systems, Information Retrieval and Analysis

Industrial Internship

Android Client for GWD Tool

Company **Samsung R&D Institute India, Bangalore**

Duration May 2017 - July 2017

Mentor **Preeti Rani Deka**

Description This was related to the Android Client for the **Samsung Gear Watch Device** (GWD) that displays the selected watch face (dial). Earlier only a few specific watch faces could be rendered on the phone, and all of that was hard coded.

Challenge Android Views did not provide support for many of the possible features in a watch face, and hence some tweaks had to be thought of.

Role My role was to create a generalized parser that can parse any general watch face XML file and render it on the phone.

Technical Skills

Programming Languages C, C++, Java, Python, SQL, SML

IDEs Eclipse, BlueJ, MATLAB, LaTeX, Android Studio

Web Development HTML, CSS, JavaScript

Awards and Scholarships

June 2018 **Commonwealth Research Scholarship**, *Nanyang Technological University*, Singapore.
Awarded the scholarship for pursuing PhD

June 2014 **JEE Advanced (IIT-JEE)**.
Secured **All India Rank 98** out of around 1.3 million students

February 2013 **Kishore Vaigyanik Protsahan Yojana (KVPY)**, *Department of Science and Technology (DST)*, Government of India.
Selected as a **Fellow** with **All India Rank 56** out of around 0.15 million candidates

September 2010 **National Talent Search Examination (NTSE)**, *National Council of Educational Research and Training (NCERT)*, India.
Selected as a **Scholar**

Languages

Bengali **Mother Tongue**

English **Academic and Professional**

Hindi **Basic**